

3.3 String formatting

Settings

Beta features

formatted string literal / f-string

A formatted string literal, or f-string, allows a programmer to create expressions that are evaluated as the program executes.

replacement field

A placeholder expression is also called a replacement field, as its value is replaced by the expression in the final output.

PARTICIPATION ACTIVITY

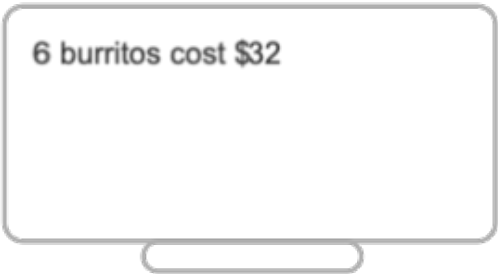
3.3.1: Creating literal strings with embedded expressions.

Start

☐ 2x speed

```
number = 6
amount = 32

print(f"{number} burritos cost ${amount}")
```



The {number} and {amount} expressions in the replacement fields are evaluated to the values of 6 and 32 in the final string.

Captions ^

- 1. The first expression, {number}, is replaced with the value of the number variable.
- 2. The second expression, {amount}, is replaced with the value of the amount variable.

Table 3.3.1: f-string examples.

Example	Output
<code>print(f"2**2=")</code>	<code>2**2=</code>
<code>two_power_two = 2**2</code> <code>print(f"{two_power_two=}")</code>	<code>two_power_two=4</code>
<code>print(f"2**2={2**4}")</code>	<code>2**2=4, 2**4=16</code>
<code>print(f"{{2**2}}")</code>	<code>{2**2}</code>
<code>print(f"{{{2**2=}}}")</code>	<code>{2**2=4}</code>

format specification

A format specification inside a replacement field allows a value's format to be customized.

presentation type

A presentation type is part of a format specification that determines how a value is presented in text form, such as an integer (4), a floating point (4.0), a fixed-point (4.000), a percentage (4%), a binary (100), etc.